

RHEINE BENTLAGE, GermanyWMO: **103060**Lat: **52.2887N**Lon: **7.3867E**Elev: **129**StdP: **14.63**Time Zone: **1.00 (EUC)**Period: **94-17**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.3	21.5	12.5	10.4	19.1	16.4	12.7	23.6	25.0	42.5	22.1	43.9	6.2	60	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	86.1	67.7	82.0	66.1	78.2	64.3	69.5	81.9	67.6	78.4	65.7	75.5	7.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	93.8	73.4	63.7	88.6	71.8	62.1	83.9	70.1	33.6	82.8	32.0	78.8	30.6	75.9	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	19.0	16.5	14.5	12.6	93.2	6.9	2.6	7.6	95.0	3.6	96.6	-0.3	98.0	-5.3	99.9	
(5)				11.8	72.9	6.9	2.4	6.9	74.6	2.9	76.0	-1.0	77.4	-6.0	79.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	36.4	38.0	42.9	49.1	56.1	61.0	65.0	64.3	58.5	51.2	43.6	38.4
	DBStd	12.02	8.20	7.48	6.22	6.70	6.47	6.00	5.83	5.49	5.17	6.22	6.43	8.12
	HDD50	1735	422	337	230	97	17	0	0	0	1	61	206	364
	HDD65	5493	886	755	684	477	285	151	75	81	206	428	641	824
	CDD50	1905	2	2	11	71	206	331	464	443	257	98	15	5
	CDD65	183	0	0	0	0	9	31	73	59	11	0	0	0
	CDH74	1867	0	0	0	20	121	327	730	563	105	1	0	0
	CDH80	554	0	0	0	1	21	83	243	188	18	0	0	0
Wind	WSAvg	6.8	7.9	7.9	7.8	6.8	6.6	6.2	6.2	5.7	5.8	6.4	6.6	7.2
Precipitation	PrecAvg	30.9	2.8	2.2	2.1	1.8	2.2	2.8	3.1	3.0	2.7	2.6	2.8	2.9
	PrecMax	41.7	5.0	5.1	5.0	3.7	4.8	6.3	6.7	8.0	5.9	8.5	5.4	6.7
	PrecMin	25.5	0.3	0.7	0.4	0.1	1.1	0.7	0.9	0.8	0.2	0.6	0.1	0.9
	PrecStd	4.8	1.3	1.2	1.3	0.8	1.0	1.3	1.4	1.7	1.4	1.5	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	56.8	66.2	77.2	83.2	88.2	91.5	89.9	82.6	71.2	61.2	56.5
		MCWB	50.8	48.6	52.7	57.9	64.6	69.4	69.3	68.2	66.6	61.9	55.9	52.5
	2%	DB	52.0	53.2	60.7	70.4	77.1	81.8	86.3	85.5	77.0	66.2	57.5	53.9
		MCWB	48.5	47.9	50.5	56.1	60.8	65.9	68.2	66.9	64.8	59.4	53.8	50.6
	5%	DB	49.7	50.6	56.1	66.1	73.3	77.4	81.9	80.5	71.8	62.9	55.2	51.9
		MCWB	46.7	46.6	48.7	54.2	59.2	63.7	66.7	66.0	61.9	57.6	51.9	50.1
	10%	DB	46.9	48.3	53.0	61.5	69.3	73.4	77.3	75.5	67.9	60.4	52.9	49.8
		MCWB	44.3	45.0	47.1	52.1	57.6	61.7	64.7	63.6	59.8	56.2	49.7	47.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.7	50.8	54.7	60.6	65.9	70.9	71.8	71.0	68.2	62.8	56.7	53.1
		MCDB	54.0	54.3	62.4	71.4	78.1	82.3	86.0	84.9	79.1	69.6	59.7	55.1
	2%	WB	49.2	48.8	51.8	57.6	63.2	67.8	69.8	68.7	65.4	60.3	54.4	52.0
		MCDB	51.6	52.0	58.2	67.2	73.8	79.0	82.1	81.2	74.6	64.9	57.3	53.2
	5%	WB	46.6	47.0	49.8	55.5	60.7	65.2	67.9	66.6	63.2	58.4	52.3	50.3
		MCDB	49.0	50.2	55.3	63.9	69.7	74.9	78.5	77.3	70.4	62.0	55.1	52.5
	10%	WB	44.4	44.9	47.7	53.1	58.4	63.0	65.9	64.8	61.0	56.5	50.0	47.2
		MCDB	46.9	48.0	52.2	60.4	67.0	71.1	75.4	73.9	66.6	60.0	52.3	49.5
Mean Daily Temperature Range	5% DB	MDBR	8.6	10.1	13.8	17.6	18.0	18.3	17.9	17.9	16.2	13.4	9.5	8.0
		MCDBR	10.4	12.4	20.7	24.9	24.8	25.5	26.3	26.0	22.8	16.4	11.5	8.0
		MCWBR	8.5	9.0	12.8	13.1	12.2	12.6	11.3	11.4	12.3	10.6	9.0	6.8
	5% WB	MCDBR	9.3	10.5	17.8	21.1	21.5	22.0	23.0	21.9	20.1	14.8	10.6	7.0
MCWBR		8.5	8.5	11.9	11.8	11.9	11.9	10.5	11.0	11.4	10.2	8.7	6.2	
Clear-Sky Solar Irradiance	taub	0.325	0.346	0.376	0.410	0.412	0.410	0.422	0.415	0.389	0.361	0.341	0.320	
	taud	2.358	2.341	2.270	2.219	2.247	2.271	2.248	2.287	2.350	2.408	2.374	2.343	
	Ebn at Noon	211	241	256	261	266	268	262	257	249	233	202	190	
	Edh at Noon	19	26	34	40	41	40	41	37	31	24	19	17	
All-Sky Solar Radiation	RadAvg	216	430	787	1276	1554	1651	1581	1329	967	564	259	169	
	RadStd	26	76	108	138	139	128	200	113	112	54	29	19	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD